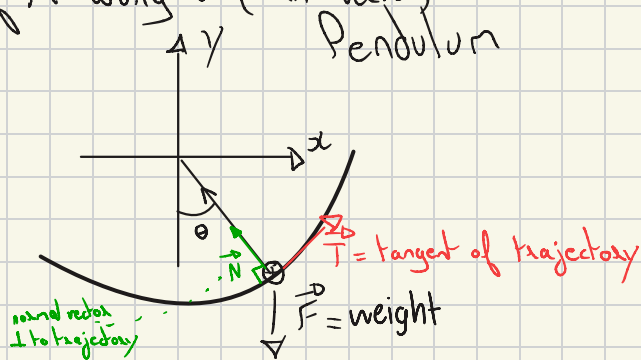
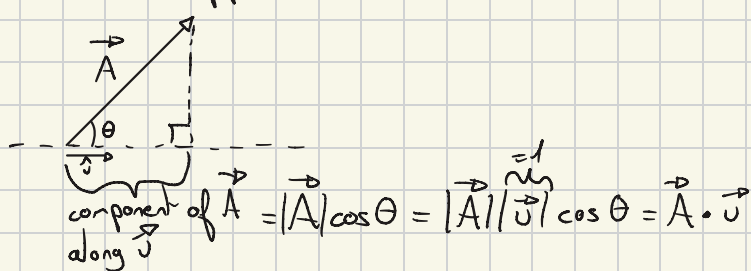


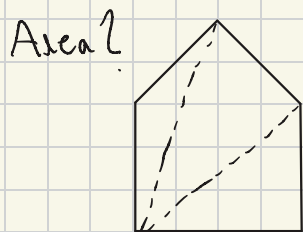
Lecture 2: Determinants; cross product

3rd application is to find components of $\vec{A}$ along $\vec{u}$ (unit vector)



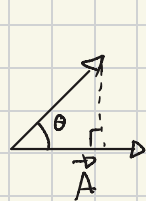
Component of $\vec{F}$ along $\vec{T}$ is what causes the pendulum to swing.

Component along $\vec{N}$ is responsible for the tension of the string.



We can divide the area of the pentagon into triangles, then add the areas of said triangles and find the area of the original pentagon.

Area of triangle = $\frac{\text{base} \times \text{height}}{2}$

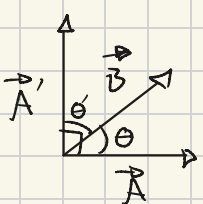


$$\text{Area} = \frac{1}{2} |\vec{A}| |\vec{B}| \sin \theta$$

Then we could find $\cos \theta$, then solve $\sin^2 \theta + \cos^2 \theta = 1$

Or: we can find $\cos \theta$ via complementary angles

$\vec{A}'$ is $\vec{A}$ rotated at a 90° angle counterclockwise



$$\theta' = \frac{\pi}{2} - \theta \quad \cos(\theta') = \sin(\theta)$$

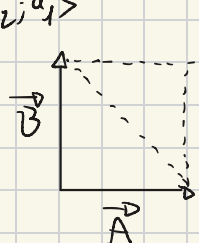
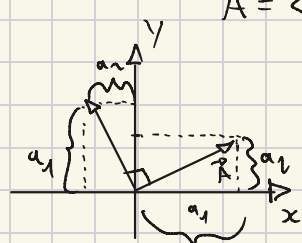
$$\vec{A}' = \langle -a_2, a_1 \rangle$$

$$\text{so, } |\vec{A}| |\vec{B}| \sin \theta \Leftrightarrow |\vec{A}'| |\vec{B}| \cos \theta'$$

$$\Leftrightarrow \vec{A}' \cdot \vec{B}$$

$$\Leftrightarrow \langle -a_2, a_1 \rangle \cdot \langle b_1, b_2 \rangle$$

$$\Leftrightarrow a_1 b_2 - a_2 b_1$$



$$\pm \text{area}(\square) = |\vec{A}| |\vec{B}| \sin \theta = \det(\vec{A}, \vec{B})$$

$$\pm \text{area}(\triangle) = \frac{1}{2} |\vec{A}| |\vec{B}| \sin \theta = \frac{1}{2} \det(\vec{A}, \vec{B})$$

$$\Leftrightarrow \det(\vec{A}, \vec{B}) \Leftrightarrow \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Determinant of $\vec{A}$ and $\vec{B} = \pm \text{area of } \square$

Determinant in space: 3 vectors: $\vec{A}, \vec{B}, \vec{C}$

Theorem: Geometrically, $\det(\vec{A}, \vec{B}, \vec{C}) = \pm \text{volume of a parallelepiped}$



$$\det(\vec{A}, \vec{B}, \vec{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Cross-product of 2 vectors in a 3D space.

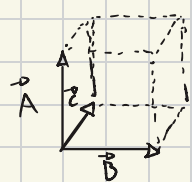
$$\text{Def: } \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{k}$$

Theorem: $* |\vec{A} \times \vec{B}| = \text{area of parallelogram}$

$* \text{dir}(\vec{A} \times \vec{B}) = \perp$ to the plane of the parallelogram



Another look at volume:



$$\text{Volume} = \text{area}(\text{base}) \times \text{height}$$

$$= |\vec{B} \times \vec{C}| \times (\vec{A} \cdot \vec{n})$$

$$= |\vec{B} \times \vec{C}| \left(\vec{A} \cdot \frac{\vec{B} \times \vec{C}}{|\vec{B} \times \vec{C}|} \right)$$

$$= \vec{A} \cdot (\vec{B} \times \vec{C}) = \det(\vec{A}, \vec{B}, \vec{C})$$

$$\vec{n} = \frac{\vec{B} \times \vec{C}}{|\vec{B} \times \vec{C}|}$$